

Kaden Downes

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Summary

Aspiring Embedded Software Engineer with hands-on experience in ARM Cortex-M firmware development, FreeRTOS, PCB design, and hardware debugging. Strong background in low-level computer hardware, communication protocols, and cross disciplinary engineering projects. Available to begin full-time July 2026. Authorized to work in the U.S. without sponsorship. Willing to relocate as needed.

Skills

- Embedded Systems
- FreeRTOS and related concepts
- PCB and RTL Design
- Communication Protocols (UART, I2C, SPI, Wi-Fi, Bluetooth)
- Hardware Debugging (J-Link, Logic Analyzers)
- **Tools:** Altium, MATLAB, Git, Azure DevOps, Vivado, Microsoft Office
- **Embedded Languages:** C/C++, Rust
- **Other Languages:** C#, Python, VHDL, Verilog, Dart, LaTeX
- **Other Skills:** Linux, Networking, Troubleshooting, System Administration

Education

Bachelor of Science in Electrical Engineering	Purdue University Fort Wayne	2022–2026
Bachelor of Science in Computer Engineering	Purdue University Fort Wayne	2022–2026

Relevant Coursework

- Embedded Microprocessors — ARM architecture, peripheral interfacing, low-level firmware development
- Real-Time Operating Systems (FreeRTOS) — task scheduling, ISR design, synchronization primitives
- Computer Architecture — pipelining, memory hierarchy, and performance optimization for constrained systems

Projects

Battery Powered Wet Floor Sensor (Senior Design) *Sponsor: Franklin Electric*
Implemented device firmware in C for an ARM Cortex-M MCU using FreeRTOS; managed tasks, ISRs, and power states. Integrated Wi-Fi module with Bluetooth provisioning and implemented TCP communication for telemetry and alerts. Built project with CMake/Ninja and managed source with Git. Designed PCB in Altium for manufacturing. Supported hardware bring-up and debugging with J-Link and logic analyzers.

Certifications

- Google IT Support Professional Certificate (v2) — Coursera, 2025

Experience

Advanced Repair Agent Geek Squad — Fort Wayne Nov 2021–Present

- Performed detailed fault analysis to determine software vs. hardware causes and carried out component level repairs on multiple device platforms.
- Developed automation scripts using PowerShell and Bash to streamline workflows.
- Closed 137 tickets every month, averaging 1.2 tickets per hour with 140% labor efficiency.
- Communicated with clients and other agents as to the status of and next steps for work orders.
- Responsibly handled sensitive customer data, including storing and transferring personal files.
- Achieved 100% customer satisfaction according to Apple repair surveys.